Limit of some special sequence.

Thm. If $p>0$, $\lim_{n\to\infty} \frac{1}{n^p}=0$

proof. Let $\varepsilon>0$ be given. Take $N\in\mathbb{N}$ s.t

$$\left(\frac{1}{\varepsilon}\right)^{\frac{1}{p}} < N \qquad \left(\Leftrightarrow \frac{1}{N^p}<\varepsilon\right)$$

Then, for all $n\geq N$,

$$\frac{1}{n^p}\leq\frac{1}{N^p}<\varepsilon.$$

Thm. If $p>0$, $\lim_{n\to\infty}\sqrt[n]{p}=1$

proof. If $p=1$, then trivial.

If $p>1$, then $x_n=\sqrt[n]{p}-1>0$. By binomial Thm,

$$1+nx_n\leq(1+x_n)^n=p.$$

Now,

$$0<x_n\leq\frac{p-1}{n}\to 0 \quad \text{as } n\to\infty$$

Thus, $\lim_{n\to\infty}\sqrt[n]{p}-1=0$.

If $p<1$, then,

$$\lim_{n\to\infty}\sqrt[n]{\frac{1}{p}}=\lim_{n\to\infty}\frac{1}{\sqrt[n]{p}}=1.$$

Thus $\lim_{n\to\infty}\sqrt[n]{p}=1$.

Thm. $\lim_{n\to\infty}\sqrt[n]{n}=1$

proof. put $x_n=\sqrt[n]{n}-1$. Then, by the binomial Thm,

$$n=(x_n+1)^n\geq\binom{n}{2}x_n^2=\frac{n(n-1)}{2}\cdot x_n^2$$

Now, $0<x_n\leq\sqrt{\frac{2n}{n(n-1)}}=\sqrt{\frac{2}{n-1}}\to 0$ as $n\to\infty$.

Corollary. For any non-zero $f(x)\in\mathbb{R}[x]$,

$$\lim_{n\to\infty}\sqrt[n]{f(n)}=1.$$

Thm. If $p>0$ and $\alpha\in\mathbb{R}$, then
$$\lim_{n\to\infty}\frac{n^\alpha}{(1+p)^n}=0.$$

Proof. Take $k\in\mathbb{Z}$ s.t. $k>\max\{\alpha,0\}$. Then, for $n\geq 2k$,
$$(1+p)^n>\binom{n}{k}p^k=\frac{n!}{(n-k)!\cdot k!}\cdot p^k>\left(\frac{n}{2}\right)^k\cdot\frac{p^k}{k!}$$
$$\Rightarrow \frac{n^\alpha}{(1+p)^n}<\frac{2^k k!}{n^k p^k}\cdot n^\alpha=\frac{2^k k!}{p^k}\cdot n^{\alpha-k}\to 0 \quad \text{as } n\to\infty, \text{ since } \alpha-k<0.$$

Thm. If $0\leq x<1$, then
$$\sum_{n=0}^{\infty}x^n=\frac{1}{1-x}$$

Proof. Since
$$(1-x)(1+x+\cdots+x^n)=1-x^{n+1}\Leftrightarrow\sum_{k=1}^{n}x^k=\frac{1-x^{n+1}}{1-x}$$
And $x^{n+1}\to 0$ as $n\to\infty$, we obtain the result.

Thm. Suppose that $\{a_n\}$ is decreasing non-negative number sequence. Then,
$$\sum_{n=1}^{\infty}a_n \text{ Converges iff } \sum_{n=1}^{\infty}2^k a_{2^k} \text{ Converges.}$$

proof. Since $\{a_n\}$ is Consisted by nonnegative terms, it suffices to find bound for partial sum.

Denote $S_n=\sum_{i=1}^{n}a_i$ and $t_k=\sum_{j=1}^{k}2^j\cdot a_{2^j}$

If $n<2^k$, then
$$\begin{aligned}S_n&=a_1+a_2+\cdots+a_n\\&\leq a_1+a_2+\cdots+a_{2^k-1}+a_{2^k}+\cdots+a_{2^{k+1}-1}\\&\leq a_1+2a_2+2^2a_{2^2}+\cdots+2^k\cdot a_{2^k}\\&=t_k\end{aligned}$$

If $n>2^k$, then
$$\begin{aligned}S_n&=a_1+a_2+\cdots+a_n\\&\geq a_1+a_2+\cdots+a_{2^{k-1}+1}+\cdots+a_{2^k}\\&\geq\frac{1}{2}a_1+a_2+2a_{2^2}+2^2\cdot a_{2^3}+\cdots+2^{k-1}\cdot a_{2^k}\\&=\frac{1}{2}t_k.\end{aligned}$$

Now, if S_n is bounded, then t_k is bounded, similarly converse.

Thm. If $p>1$, then $\sum_{n=2}^{\infty} \frac{1}{n(\log n)^p}$ Converges.

If $p \le 1$, then diverges.

Proof. Since $\sum_{n=2}^{\infty} \frac{1}{n(\log n)^p}$ Converges if and only if

$$\sum_{k=1}^{\infty} 2^k \cdot \frac{1}{2^k \cdot (\log 2^k)^p} = \sum_{k=1}^{\infty} \frac{1}{k^p \cdot (\log 2)^p} = \frac{1}{(\log 2)^p} \cdot \sum_{k=1}^{\infty} \frac{1}{k^p}$$

we obtain the result.

Thm. For any $c \in \mathbb{R} \setminus \mathbb{Z}$, $\sum_{n=1}^{\infty} \frac{1}{n+c}$ diverges.

Proof. Take $k \in \mathbb{N}$ s.t. $k > c+1$ Then,

$$\sum_{n=1}^{\infty} \frac{1}{n} \le \sum_{n=k}^{\infty} \frac{1}{n} \le \sum_{n=1}^{\infty} \frac{1}{n+c}$$

Def. $e \overset{\text{def}}{=} \sum_{n=0}^{\infty} \frac{1}{n!}$.

Since $S_n = \sum_{k=0}^{n} \frac{1}{k!} = 1 + 1 + \frac{1}{1\cdot 2} + \frac{1}{1\cdot 2\cdot 3} + \cdots + \frac{1}{1\cdot 2\cdots n} \le 1 + \sum_{k=0}^{n} \frac{1}{2^k} \to 3$, well-defined.

Thm. $\lim_{n\to\infty} \left(1+\frac{1}{n}\right)^n = e$.

Proof. put $S_n = \sum_{k=0}^{n} \frac{1}{k!}$, $t_n = \left(1+\frac{1}{n}\right)^n$.

By the binomial theorem,

$$t_n = \left(1+\frac{1}{n}\right)^n = \sum_{k=0}^{n} \frac{n!}{(n-k)!\cdot k!} \cdot \frac{1}{n^k} = \sum_{k=0}^{n} \frac{1}{k!} \cdot \frac{n\cdot(n-1)\cdots(n-k+1)}{n^k}$$

$$= \sum_{k=0}^{n} \frac{1}{k!} \cdot \left(1-\frac{1}{n}\right)\left(1-\frac{2}{n}\right)\cdots\left(1-\frac{k-1}{n}\right)$$

$$\le \sum_{k=0}^{n} \frac{1}{k!} = S_n.$$

$n \ge m$

$\frac{1}{n} \le \frac{1}{m}$

$1-\frac{1}{n} \ge 1-\frac{1}{m}$

So that, $\limsup_{n\to\infty} t_n \le \limsup_{n\to\infty} S_n = \lim_{n\to\infty} S_n = e$.

Meanwhile, if $n \ge m$, then

$$t_n = \sum_{k=0}^{n} \frac{1}{k!} \cdot \left(1-\frac{1}{n}\right)\left(1-\frac{2}{n}\right)\cdots\left(1-\frac{k-1}{n}\right)$$

$$\ge \sum_{k=0}^{m} \frac{1}{k!} \cdot \left(1-\frac{1}{n}\right)\left(1-\frac{2}{n}\right)\cdots\left(1-\frac{k-1}{n}\right) \quad \text{(Note: this is not } t_m\text{)}$$

$$\ge \sum_{k=0}^{m} \frac{1}{k!} = S_m$$

Therefore, for any $m \in \mathbb{N}$, $\liminf_{n\to\infty} t_n \ge S_m$

Now, $\liminf_{n\to\infty} t_n \ge \liminf_{m\to\infty} S_m = \lim_{m\to\infty} S_m = e$.

Consequently,

$$e \le \liminf_{n\to\infty} t_n \le \limsup_{n\to\infty} t_n \le e.$$

Note: for all $n \in \mathbb{N}$,

$$e - S_n = \sum_{k=n+1}^{\infty} \frac{1}{k!} = \frac{1}{(n+1)!} \cdot \sum_{k=0}^{\infty} \frac{1}{k!} < \frac{1}{(n+1)!} \sum_{k=0}^{\infty} \frac{1}{(1+n)^k} = \frac{1}{n!\cdot n}$$

Thm. e is irrational.

Proof. Suppose that e is rational. Since e is positive, $e = \frac{p}{q}$ for some $p, q \in \mathbb{N}$.

Denote $S_n = \sum_{k=0}^{n} \frac{1}{k!}$. Then, by approximation of e,

$$0 < q!(e - S_q) < \frac{1}{q}.$$

By assumption, $q!e = p \cdot (q-1)!$ is an integer, and

$$q! \cdot S_q = \sum_{k=0}^{q} \frac{q!}{k!}$$

is integer, being sum of integers.

Now, $q!(e - S_q)$ is an integer between 0 and 1. It is contradiction.

⌈Prop. Let $S_1=\sqrt{2}$, and $S_{n+1}=\sqrt{2+\sqrt{S_n}}$ $(n\geq 1)$.

Then, S_n Converges,

Proof. At first, $\sqrt{2}\leq S_1=\sqrt{2}<2$.

Suppose that for $n\in\mathbb{N}$, $\sqrt{2}<S_n<2$. Then,

$$\sqrt{2}<S_{n+1}=\sqrt{2+\sqrt{S_n}}<\sqrt{2+\sqrt{4}}=2$$

Thus, $\{S_n\}$ lies in $(\sqrt{2},2)$.

And, $S_2=\sqrt{2+\sqrt{S_1}}=\sqrt{2+\sqrt{\sqrt{2}}}>\sqrt{2}=S_1$.

Suppose that for $n\geq 2$, $S_n>S_{n-1}$

Then, $S_{n+1}=\sqrt{2+\sqrt{S_n}}>\sqrt{2+\sqrt{S_{n-1}}}=S_n$

Now, $\{S_n\}$ is increasing bounded sequence, thus converges.

⌈Prop. If $\{a_n\}$ is non-negative and $\sum a_n$ Converges, then $\sum\frac{\sqrt{a_n}}{n}$ Converges,

Proof. observe that: for all $n\in\mathbb{N}$,

$$0\leq(n\sqrt{a_n}-1)^2=n^2a_n-2n\sqrt{a_n}+1 \iff \frac{\sqrt{a_n}}{n}\leq\frac{1}{2}\cdot\left(a_n+\frac{1}{n^2}\right)$$

or using Cauchy-Schwarz inequality:

$$\sum_{n=1}^{N}\frac{\sqrt{a_n}}{n}\leq\sqrt{\sum_{n=1}^{N}a_n}\cdot\sqrt{\sum_{n=1}^{N}\frac{1}{n^2}}.$$

⌈Lemma. Cauchy-Schwarz inequality

$$\sum_{i=1}^{n}|a_ib_i|\leq\sqrt{\sum_{i=1}^{n}a_i^2}\cdot\sqrt{\sum_{i=1}^{n}b_i^2}$$

Prop. Suppose $\sum a_n$ converges absolutely.

Then, the following series are absolutely convergent:

$$\sum_{n=1}^{\infty} a_n^2, \qquad \sum_{n=1}^{\infty} \frac{a_n}{1+a_n} \ (a_n \neq -1), \qquad \sum_{n=1}^{\infty} \frac{a_n^2}{1+a_n^2}$$

Proof. Since $\sum a_n$ converges absolutely, there exists $N_1 \in \mathbb{N}$ s.t.

$$n \geq N_1 \Rightarrow |a_n| < \frac{1}{2}$$

Now since $x^2 \leq x$ for $0 \leq x \leq 1$, $m \geq N_1$ implies

$$\sum_{n=1}^{m} |a_n^2| \leq \sum_{n=1}^{N_1} |a_n^2| + \sum_{n=N_1}^{m} |a_n^2| \leq \sum_{n=1}^{N_1} |a_n^2| + \sum_{n=N_1}^{m} |a_n| \leq \sum_{n=1}^{N_1} + \sum_{n=N_1}^{\infty} |a_n|$$

This implies that $\sum^{\infty} \frac{a_n^2}{1+a_n^2}$ converges directly:

$$\sum_{n=1}^{\infty} \left| \frac{a_n^2}{1+a_n^2} \right| = \sum_{n=1}^{\infty} \frac{a_n^2}{1+a_n^2} \leq \sum_{n=1}^{\infty} a_n^2.$$

Meanwhile, $|a_n| < \frac{1}{2}$ implies

$$\frac{1}{2} < |1+a_n| < \frac{3}{2} \iff \frac{2}{3} < \frac{1}{|1+a_n|} < 2$$

That is, for $n \geq N_1$, $\left| \frac{a_n}{1+a_n} \right| \leq 2|a_n|$, thus $\sum_{n=1}^{\infty} \frac{a_n}{1+a_n}$ converges absolutely.

「Prop. If $\sum a_n$ Converges and $\{b_n\}$ is monotonic and bounded, then $\sum a_n b_n$ Converges.

Proof. Since $\{b_n\}$ is monotonic and bounded, it is convergent. Let $b = \lim_{n\to\infty} b_n$.

Suppose that $\{b_n\}$ is decreasing. Then, for all $n \in \mathbb{N}$, $b_n \geq b$.

Now, the sequence $\{b_n - b\}$ is monotonically decreasing and $b_n - b \to 0$ as $n \to \infty$.

Therefore, $\sum a_n(b_n - b)$ Converges.

「Counterexample if $\{b_n\}$ Converges but not monotone.

$\sum (-1)^n \frac{1}{\sqrt{n}}$ Converges, and $b_n = (-1)^n \frac{1}{\sqrt{n}}$ Converges.

But $\sum \frac{1}{n}$ not converges.